Invisible No More

Common sense visualization with ggplot

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Why do we visualize data?

Why do we visualize data?

When we visualize data,
we do this to **communicate** it

Visual Communication

Thinking of plotting as a form of communication allows us to draw an analogy to languages.
A formal language can be fully described by specifying their *grammar* and *lexicon*.

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Formally defining a grammar allows us to

Grammar of Graphics

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- Describe plots systematically
- Understand plots we see
- Flexibly modify existing or invent new plots
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It does not provide advice for making beautiful or useful plots!

Grammar of Graphics: Components

In the (layered) grammar of graphics, a plot consists of:

- Data, (Transformations)
- Aesthetics/Mappings
- Geometries
- Stats
- Scales
- Coordinates

Grammar of Graphics: Components

In the (layered) grammar of graphics, a plot consists of:

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- Geometries
- Stats
- Scales
- Coordinates

These components can be swapped or altered individually, and composed to make a plot. In practice, many of these have sensible default values.

Data

- *Base* of the plot
- A plot may integrate multiple datasets

Stat(istics)

- Usually aggregates data
- May output new/other values
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- Examples:
  - Identity
  - Count
  - Mean/Median
  - Smoothing, Regressions

Geom(etry)

- Graphical primitives the plot is constructed from
- May have different modifiable attributes
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- Graphical primitives the plot is constructed from
- May have different modifiable attributes
- Has a default stat
- Examples:
  - Points
  - Line
  - Bars
  - Boxes

Scales

- Describes the *how* of mapping variables to aesthetics
- Controls of guides
- Can be discrete/continuous, in number/colour/other spaces

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- Controls of guides
- Can be discrete/continuous, in number/colour/other spaces
- Examples:
    - Linear
    - Log
    - Various colourscales
    - Binning

Coordinates

- Maps scale values onto plotting coordinates
- Examples:

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- Examples:
  - Cartesian, 2D/3D
  - Polar
- Commonly cartesian, polar, 3D

Aesthetics/Mappings

- **Which variable gets plotted how after which transformation, and how?**
- *Glue* of ggplot

How do these components combine to make a plot?

Assembling a ggplot

- A plot may be multiple layers
- Each layer has one data, stat, geom and aesthetics
- These pass information along to render the plot

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- A plot may be multiple layers
- Each layer has one data, stat, geom and aesthetics
- These pass information along to render the plot
- In practice, defaults often work well

Let's have a look at how this looks like in some plots!

`ggplot2`: Implementing a grammar of graphics

Now, let’s make some plots ourselves!